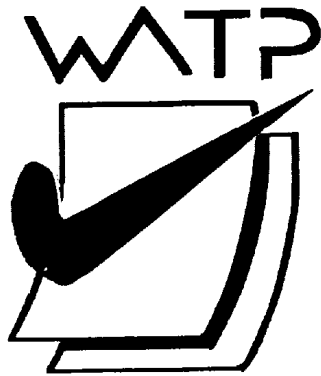




**Semester One  
Examination 2017  
Question/Answer booklet**

**MATHEMATICS  
APPLICATIONS UNIT 1**

**Section One:  
Calculator-free**

Student Name: \_\_\_\_\_

Teacher's Name: \_\_\_\_\_

**Time allowed for this section**

Reading time before commencing work: five minutes  
Working time for paper: fifty minutes

**Material required/recommended for this section**

**To be provided by the supervisor**

This Question/Answer booklet  
Formula Sheet

**To be provided by the candidate**

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,  
correction tape/fluid, eraser, ruler, highlighters

Special Items: nil

**Important note to candidates**

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

## Structure of this paper

|                                        | Number of questions available | Number of questions to be attempted | Working time (minutes) | Marks available | Percentage of exam |
|----------------------------------------|-------------------------------|-------------------------------------|------------------------|-----------------|--------------------|
| <b>Section One<br/>Calculator—free</b> | <b>13</b>                     | <b>13</b>                           | <b>50</b>              | <b>50</b>       | <b>35</b>          |
| Section Two<br>Calculator—assumed      | 15                            | 15                                  | 100                    | 100             | 65                 |
| Total marks                            |                               |                                     |                        |                 | 100                |

## Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2017*. Sitting this examination implies that you agree to abide by these rules.

2. Answer the questions according to the following instructions.

**Show all your working clearly.** Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

It is recommended that you **do not use pencil**, except in diagrams.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
  - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
  - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.
5. The Formula Sheet is **not** handed in with your Question/Answer Booklet.

**Section One: Calculator–free****35% (50 marks)**

This section has **thirteen (13)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Working time: 50 minutes

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**Question 1 (4 marks)**

Martha receives a tax refund of \$8 000.

She allocates half for a holiday, 30% to repay a loan and she puts the remainder into her savings account.

- (a) How much money does she put into her savings account? (2 marks)

Her tax refund was exactly 10% of her salary for the year.

- (b) How much was her salary? (2 marks)

**Question 2 (2 marks)**

The difference of two squares is factorised below.

$$x^2 - y^2 = (x + y)(x - y)$$

Use the difference of two squares formula to fully evaluate  $17.1^2 - 16.9^2$  (2 marks)

**Question 3 (3 marks)**

Val takes her elderly mother to a concert.

Val pays adult fare, while her mother's fare is concession.

Val's ticket costs \$20, and the two tickets have a combined cost of \$35.

By what percentage is the adult fare discounted for concession? (3 marks)

**Question 4 (3 marks)**

2% of a special number is \$6.30.

- (a) How much is 1% of that special number? (1 mark)
- (b) How much is 10% of that special number? (1 mark)
- (c) What is the special number? (1 mark)

**Question 5 (4 marks)**

The ratio of the volumes of two similar solid figures is 5:3.

The volume of the larger figure is 1 000 cm<sup>3</sup>.

- (a) Determine the volume of the smaller figure. (2 marks)
- (b) If the figures are cubes, determine the length of each side of the larger cube. (2 marks)

**Question 6 (2 marks)**

Calculate the simple interest on \$1000 for 3 years at 7% p.a.

(2 marks)

**Question 7 (5 marks)**

Agatha invests some money with a bank. She earns interest, compounded monthly. The final balance in her account is given by the formula below.

$$A = 5000(1.01)^{36}$$

State the value of:

(a) the principal.

(1 mark)

(b) the annual interest rate.

(2 marks)

(c) the number of years she left her money to grow.

(2 marks)

**Question 8 (5 marks)**

If  $A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 3 & 0 \\ 3 & 2 & 0 \end{bmatrix}$ , determine each of the following.

(a)  $A - B$  (1 mark)

(b)  $A - A$  (1 mark)

(c)  $-2B$  (1 mark)

(d)  $AB$  (2 marks)

**Question 9 (6 marks)**

Harry is entering into a share portfolio investment.

In Year 1, he invests in Company A (10 000 shares), Company B (8 000) and Company C (5 000).

In Year 2, he invests in Company A (8 000 shares), Company B (5 000) and Company C (10 000).

(a) Write down a  $2 \times 3$  matrix showing his investments over the two years. (2 marks)

(b) Write down a  $1 \times 3$  matrix showing his total holding in each company after 2 years. (1 mark)

Some years later, Company A shares have a value of \$2 each, Company B shares have a value of \$3 each and Company C shares have a value of \$4 each.

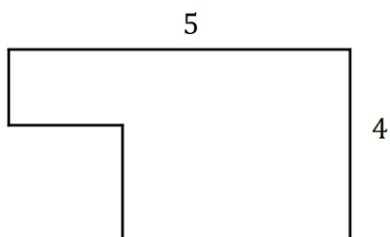
(c) Use matrix multiplication to determine the total value of his portfolio. (3 marks)

**Question 10 (5 marks)**

The following figures are not necessarily congruent. Units are in cm.

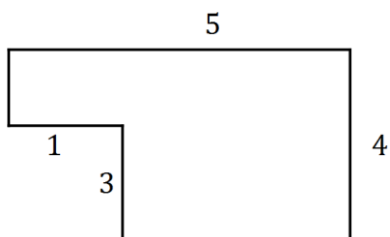
- (a) Find the perimeter of the figure shown.

(2 marks)



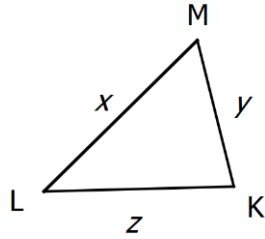
- (b) Find the area of the figure shown.

(3 marks)



**Question 11 (5 marks)**

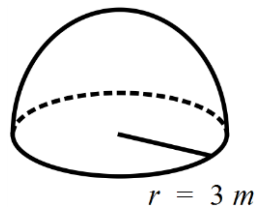
Consider triangle MLK which is not drawn to scale.  $x$ ,  $y$  and  $z$  are lengths in cm.



- (a) If angle K is a right angle, state the relationship between the three sides. (1 mark)
- (b) If angle M is a right angle, state the relationship between the three sides. (1 mark)
- (c) If  $x = 5$ ,  $y = 6$  and  $z = 7$ , which angle would you test to see whether it is a right angle? Perform the test and state your conclusion. (3 marks)

**Question 12 (4 marks)**

This hemisphere includes the base.



For the diagram above, calculate the:

(a) surface area in terms of  $\pi$ . (2 marks)

(b) volume. (2 marks)

**Question 13 (2 marks)**

Circle the item(s) which would be appropriately measured in square metres. (2 marks)

- (a) A new white board in Ben's classroom.
- (b) The distance around the new Perth stadium.
- (c) The floor area of a basketball court.
- (d) The capacity of Sydney harbour.

**End of Questions**

**Additional working space**

Question number(s): .....

**Additional working space**

Question number(s): .....

**Additional working space**

Question number(s): .....

WATP acknowledges the permission of School Curriculum and Assessment Authority in providing instructions to students.